

Centre Number	Candidate Number	Candidate Name
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NAMIBIA SENIOR SECONDARY CERTIFICATE

GEOGRAPHY ORDINARY LEVEL

6137/3

PAPER 3

2 hours

Marks 60

2024

Additional Materials: Non-programmable calculator
 Protractor
 Ruler

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Candidates answer on the Question Paper in the spaces provided.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- Do not write in the margin *For Examiner's Use*.
- Answer **all** questions.
- All working must be clearly shown.
- Sketch maps and diagrams should be drawn whenever they serve to illustrate an answer.
- The number of marks is given in brackets [] at the end of each question or part question.
- You may use a non-programmable calculator.

<i>For Examiner's Use</i>			
	Marks	Marker	Checker
Question 1			
Question 2			
Total			

This document consists of **12** printed pages.



Republic of Namibia
MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 A group of Geography learners were planning to study beach processes in Lüderitz, a coastal town. They wanted to establish the profile of the beach by measuring the angle of the beach slope. Also, they intended to investigate how the size of pebbles on the beach varied with distance from the sea.

(a) (i) A beach is a depositional landform.

Name another depositional landform that can be found on the coast.

..... [1]

(ii) Before they began their fieldwork, their teacher reminded them about safety near the sea.

Explain why the teacher asked them to
check the weather forecast.

..... [1]
.....

take a mobile phone with them.

..... [1]
.....

(b) When the learners arrived at the beach, they divided themselves into groups of three and went onto the beach to begin measuring the beach's angle to establish the beach profile. Their teacher had given each group a clinometer, two ranging poles and a tape measure.

(i) Describe how each group used their equipment to establish the beach profile.

..... [3]
.....
.....
.....
.....
.....

(ii) Describe **two** problems the group of learners may have had in ensuring their measurements were accurate.

1

.....

2

..... [2]

- (iii) To measure the size of beach material, the learners picked up a pebble every metre along their beach profile line.

Explain **one** disadvantage of this method of sampling and how it could be improved.

Disadvantage

..... [1]

Improvement

..... [1]

- (c) The results of the learners' measurements are shown in Table 1 and the results are plotted in Fig. 1.

Table 1
Results of learners' measurements

Distance from the back of the beach (m)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Difference in height between ranging poles (m)	0.0	0.2	0.4	0.5	0.5	0.6	0.6	1.0	1.1	1.2	1.2	1.6	1.7	1.8	1.8	1.9
Size (length) of pebble (cm)	18	15	20	14	13	11	11	10	2	9	9	10	5	1	1	2

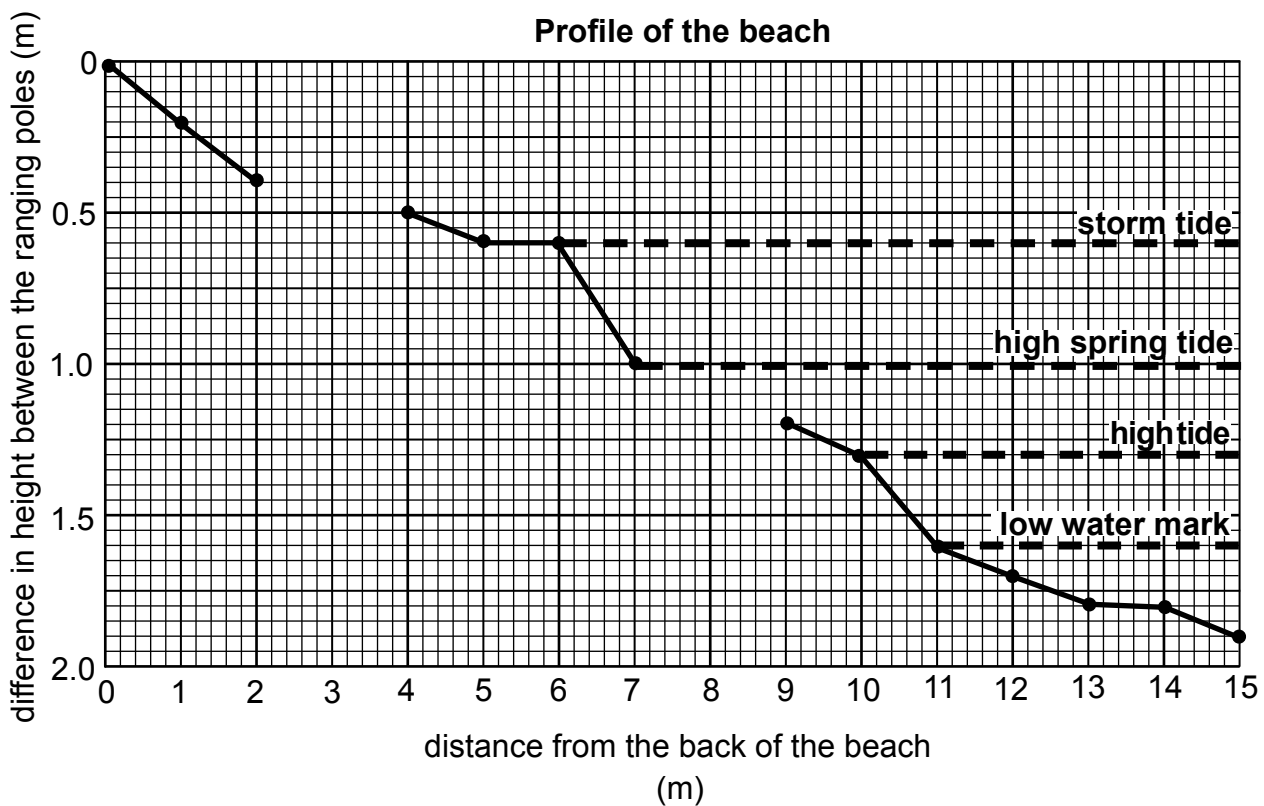


Fig. 1

- (i) Use the results in Table 1 to complete the profile of the beach on Fig. 1. [2]

(d) The learners used their results to test the following hypotheses:

Hypothesis 1: *The profile of the beach will be similar to the textbook example of a typical beach.*

Hypothesis 2: *Pebbles become longer as distance from the back of the beach increases.*

The textbook example of a beach profile is shown in Fig. 2.

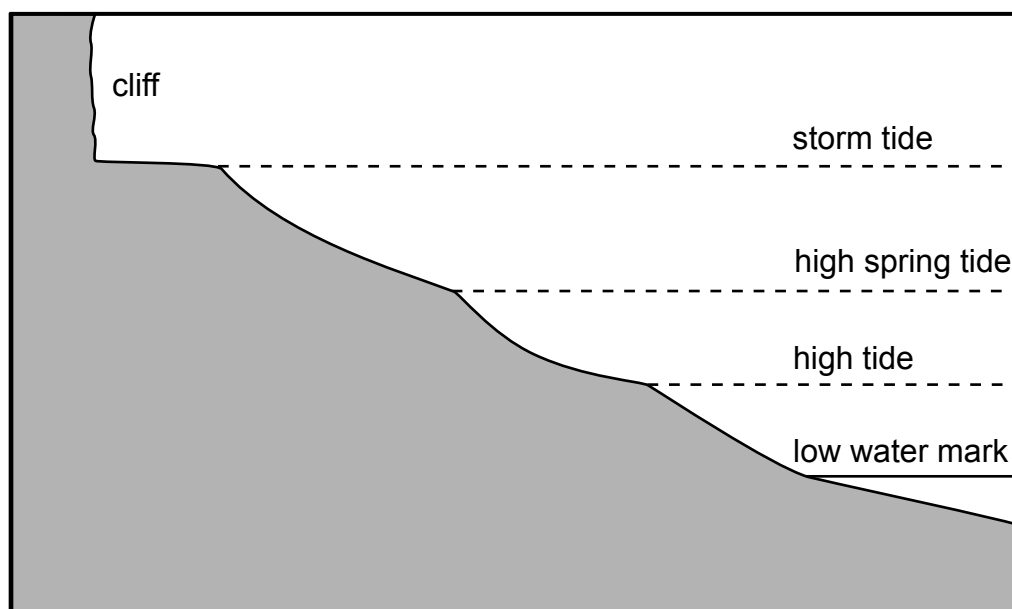


Fig. 2

- (i) What conclusion would the learners make about **Hypothesis 1:** *The profile of the beach will be similar to the textbook example of a typical beach?* Support your answer with evidence from Fig. 1 and Fig. 2.

.....

.....

.....

.....

.....

[3]

- (ii) Use the results in Table 1 to plot the pebble size measurements at 7m and 15m from the back of the beach on Fig. 3.

[2]

Pebble size across the beach

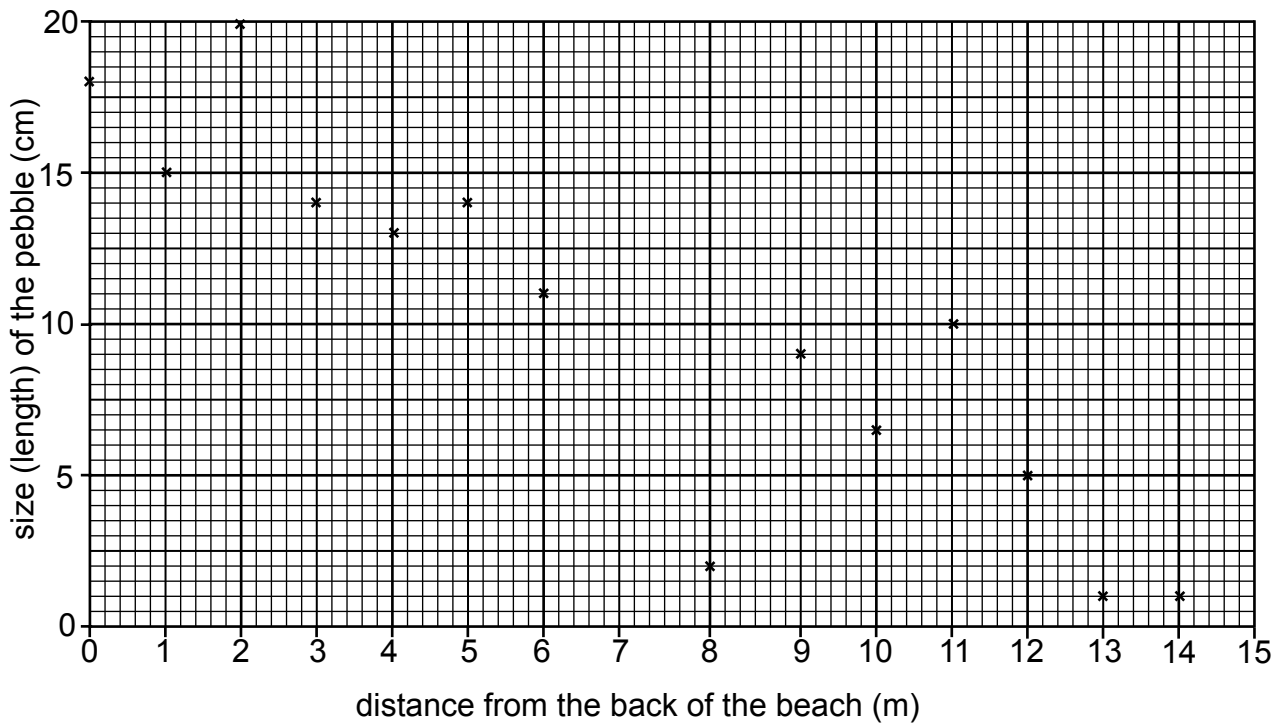


Fig. 3

- (iii) What conclusion would the learners make about **Hypothesis 2**: *Pebbles become longer as distance from the back of the beach increases*? Support your answer with evidence from Table 1 and Fig. 3.

.....

.....

.....

.....

.....

[3]

- (iv) Describe **three** reasons why the size of beach material varies across the beach shown in Fig. 3.

1

.....

2

.....

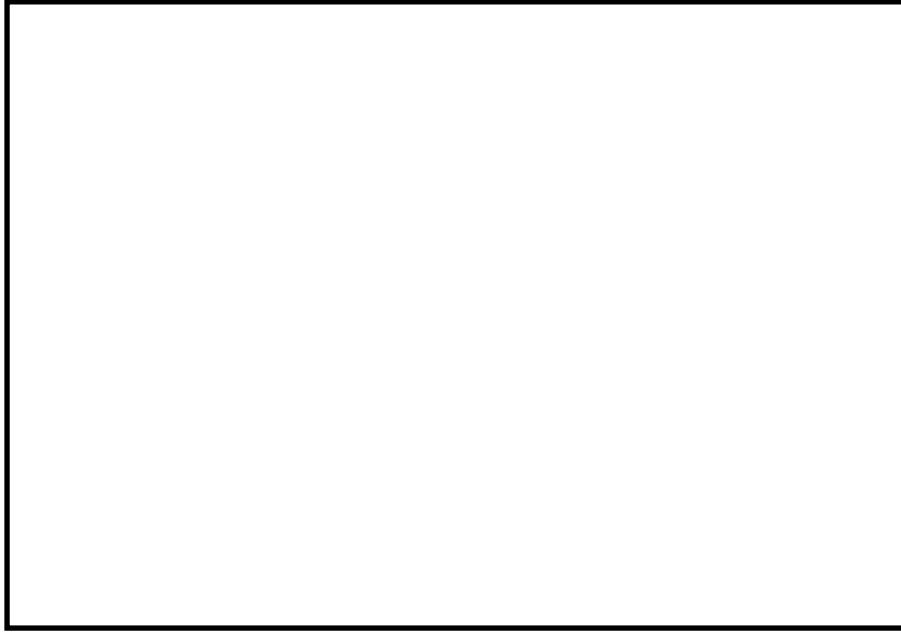
3

.....

[3]

- (e) (i) Draw a labelled diagram to describe how waves may cause longshore drift on a beach.

[4]



- (ii) Describe a fieldwork the group of learners could have carried out to determine the direction of longshore drift.

.....

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.....

.....

[3]

[30]

- 2 Another class of Geography learners in Lüderitz wanted to do an investigation of the Central Business District (CBD) of their town. The learners decided on the following hypotheses:

Hypothesis 1: *Pedestrian flow decreases away from the centre of the CBD.*

Hypothesis 2: *The tallest buildings are located in the CBD.*

They wanted to decide where the boundary of the CBD ends. They learned in class that this is “delimiting the CBD”.

- (a) (i) Apart from the high number of pedestrians and multi-storey buildings, describe **two** other characteristics expected of a town's CBD.

1

.....

2

.....

[2]

- (ii) Explain **two** reasons for the variations of building heights found around a town.

1

.....

2

.....

[2]

- (b) The learners decided to do their pedestrian counts, counting the number of pedestrians around the CBD at 100 metres, 200 metres and 300 metres distances from the town's central point, for 10 minutes at 5 different times during the day.

- (i) State **three** safety considerations learners should consider when conducting a pedestrian count.

1

2

3

[3]

- (ii) Suggest **two** reasons why learners used this method for doing the pedestrian count.

1

.....

2

.....

[2]

- (c) Study the results of the three survey sites on Independence Street which are shown in Table 2. A copy of the recording sheet used by learners is shown in Fig. 4

Table 2
Results of survey sites on Independence Street.

Distance from central point (m)	Time					Total number of pedestrians
	08:00-08:10	10:30-10:40	13:00-13:10	15:30-15:40	18:00-18:10	
100	30	25	28	16	12	111
200	17	12	11	6	5	51
300	12	6	7	4	3	32

Recording sheet

Street name:	Independence Street
Distance from central point:	200 m
Time of survey:	08:00 to 08:10
Tally	Tally Total

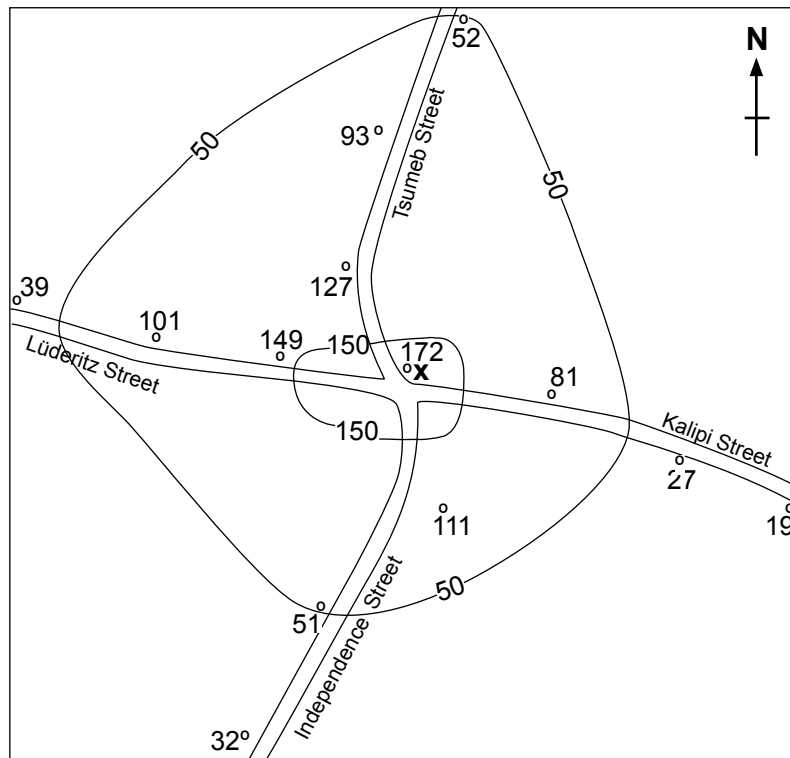
Fig. 4

- (i) Use Table 2 to complete the recording sheet by inserting the correct tally marks and tally total in Fig. 4. [2]
- (ii) Calculate the total number of pedestrians surveyed by the learners on Independence Street.

..... [1]

- (d) The results of the pedestrian count are shown in Fig. 5. Some isolines have been drawn on the map to show pedestrian flow.

Results of the pedestrian count



Key

- survey site and total number of pedestrians recorded
 - x central point
 - ~50~ isoline
- 0 100
m

Fig. 5

- (i) On Fig. 5, draw the isoline to show 100 pedestrians. [2]
- (ii) On Fig. 5, shade the area where the number of pedestrians recorded is over 150 pedestrians. [1]
- (iii) Describe the distribution of pedestrians shown in Fig. 5 for the town. Support your answer with evidence from Table 2 and Fig. 5.

.....

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.....

[3]

- (iv) What conclusion would the learners make to **Hypothesis 1**:
Pedestrian flow decreases away from the centre of the CBD?
Support your answer with evidence from Table 2.

.....

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.....

.....

[3]

- (e) To collect data about the building height, the learners counted the number of storeys observed in buildings around the central point of the town. At each road junction, they selected 10 buildings and recorded the number of storeys for each, before taking an average for each site. Table 3 shows the results for one site.

Table 3
Results for site one

Building	1	2	3	4	5	6	7	8	9	10
Number of storeys	3	3	5	2	3	4	2	4	3	3

To draw a conclusion about **Hypothesis 2: The tallest buildings are located in the CBD**, the learners decided to use the 150 pedestrian isoline to mark the edge of the CBD. The isoline is drawn on Fig. 6 which also shows the average number of storeys for each road junction around the central point.

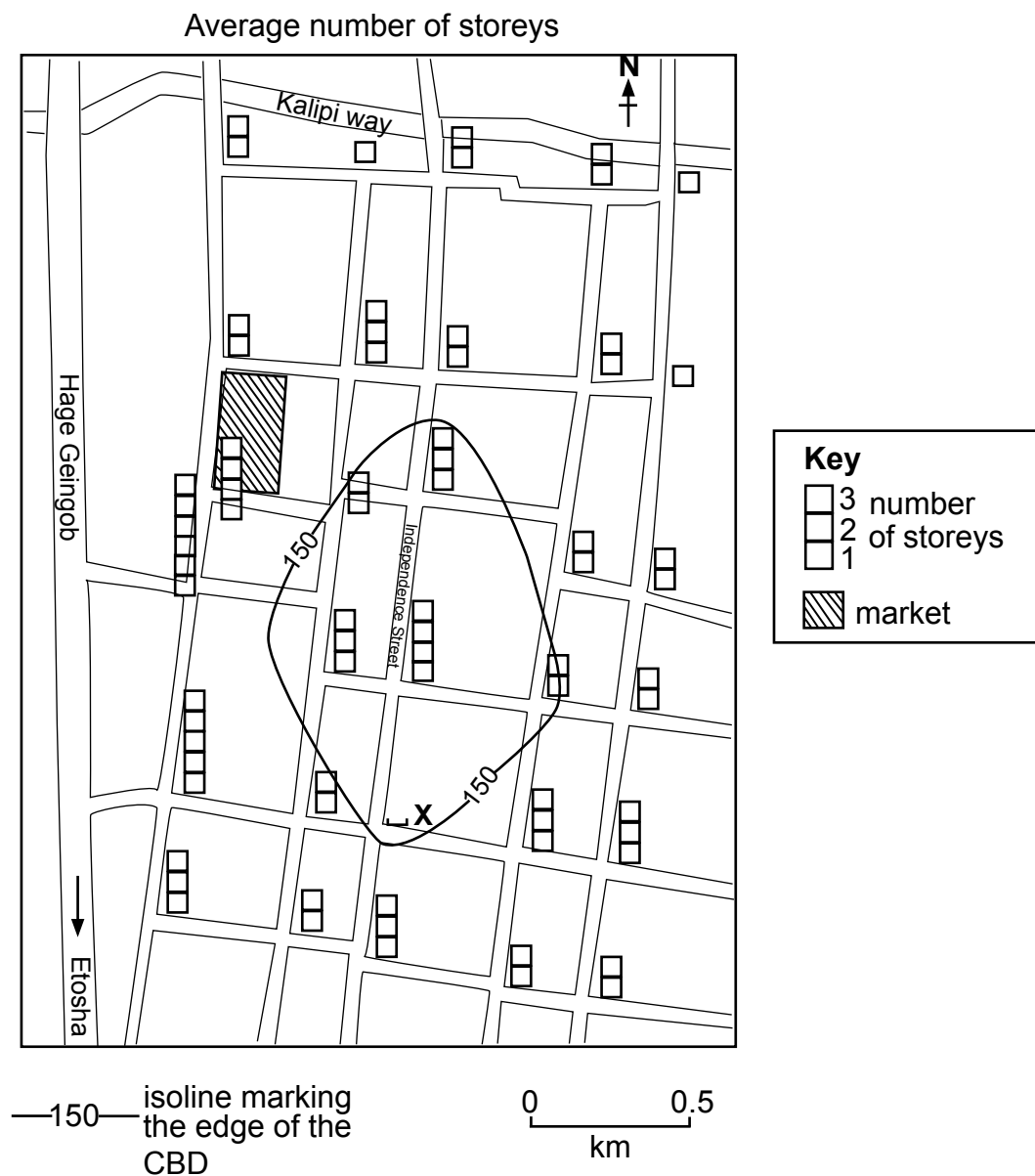


Fig. 6

- (i) Using Table 3, calculate the average number of storeys at this site. Round off your answer to a whole number.

[1]

- (ii) Suggest **one** possible problem, of using the number of storeys to estimate the height of a building.

[1]

- (iii) Describe another method for determining the height of a building, apart from counting the number of storeys.

[1]

- (iv) What conclusion would the learners make to **Hypothesis 2: *The tallest buildings are located in the CBD?*** Support your answer with evidence from Fig. 6.

[3]

- (f) One group of learners wanted to find out more about how urban function varied across the town.

Describe a method they could use to do this.

[3]

[30]

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